[Random]

Spontaneous Emmision

- Process where an atom in an excited states drops down and release a photon

$$\boxed{\text{No. of atoms in the excited state, } N(t) = N_0 e^{-t/\tau}}$$

Life time of excited state

atom in excited state at $t = 0$

When $t = \tau$,

$N(\tau) = (e^{-1}) N_0 = 0.37 N_0$

This means after time of τ, there are only 37% of excited atoms left

[63% of excited atoms have dropped down]

Take Note,

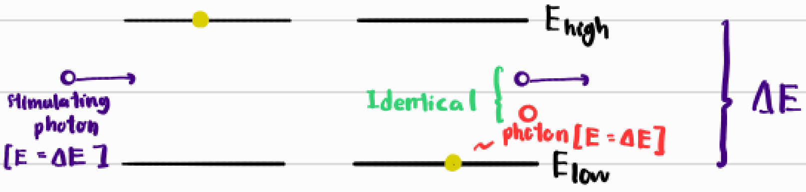
As $\tau \uparrow$, $t/\tau \downarrow$, $-t/\tau \uparrow$, $e^{-t/\tau} \downarrow$

∴ Takes a longer time to get "demoted"

If τ is large, the atom is in a metastable state

Stimulated Emmis. [SE]

SE happens when a excited atom is triggered by a passing photon (stimulating photon) with an Energy same as the ΔE btwn this excited atom and the "demoted" atom



These 2 identical photons then act as stimulating photons causing more stimulated emmision to occur. The result is tonnes of identical photons will be emmited. This is a LASER

L	A	S	E	R
Light	Amplification	Stimulated	Emmision	Radiation

LASER Challenges

① Spontaneous Emmision might happen b4 stimulated emmision

- Laser rely on tonnes of identical photons coming out at the same time, same direc. [In phase] Since human eyes can't pick up on 1 photon.

Solution : Use a metastable atom. This allows the stimulating photon enough time to cause stimulating emmision, causing a coherant laser beam

C
All photons have the same, f, λ, polarisation and phase

② At room temp, majority of atoms are not excited. These not excited atom can absorb the stimulating photon. Thus majority of the stimulating photon will be absorbed

Solution : We excite the atoms first b4 emmiting the stimulating photon.

[Population inversion]

Schrodinger Eqn [Part 2]

Wave function
[Contains info of where the particle might be]

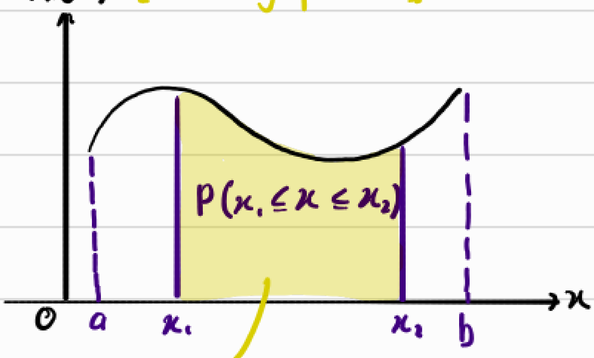
$$\boxed{E_n \psi_n(x) = \underbrace{-\frac{\hbar^2}{2m}\frac{d^2\psi_n(x)}{dx^2}}_{KE} + \underbrace{U(x)\,\psi_n(x)}_{PE}}$$ [Universal eqn]

Energy lvls the e^- is allowed to have [Allowed energies]

Bonn intepretation

Probability Density function for position of e^- [PDF_{e^-}], $\boxed{P_n(x) = \psi_n^2(x)}$

$P_n(x)$ [Probability per x]



$\int_{x_1}^{x_2} P_n(x)\, dx$

Do take note

① At position x_1, $\int_{x_1}^{x_1} P_n(x)\, dx = 0$

∴ The probability of an e^- being at a specific location is zero

Area of entire PDF must be 1

② At anywhere, $\int_a^b P_n(x)\, dx = 1$ [$\int_0^\infty P_n(x)\, dx = 1$]

∴ The e^- is definately somewhere

Heisenburgs Uncertainty Principle

$$\boxed{(\Delta x)(\Delta p) \geq \frac{h}{4\pi}}$$

Uncertainty of e^- position

Uncertainty of e^- momentum

This means $\Delta x \neq 0$, $\Delta p \neq 0$.

This makes sense because we can't precisely know an e^- position ($\int_{x_1}^{x_2} P_n(x)\,dx = 0$) or movement, we only can get the probability from Scrodinger Eqn.

De-Brogiles eqn

$$\boxed{\text{Momentum, } p = \frac{h}{\lambda}}$$

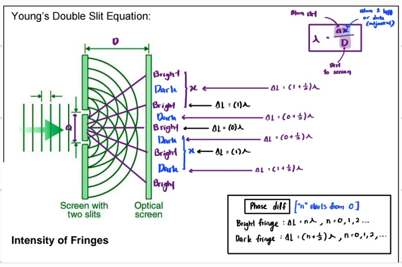
Since we know, $p = mv$, $KE = \frac{1}{2}mv^2$,

we can get $\boxed{p = \sqrt{2m(KE)}}$

$$\therefore \sqrt{2m(KE)} = \frac{h}{\lambda}$$

$$\boxed{\lambda = \frac{h}{\sqrt{2m(KE)}}}$$

Since particles act as a wave, with a wavelength λ.



This formula applies,

$$\boxed{\lambda = \frac{ax}{D}}$$